

General Features

The i213 is an intelligent high precision analog output module for the ionia™ system. The module has 8 analog outputs grouped into two galvanically isolated sections, each with its own reference ground.

Each channel can be individually selected as voltage or current outputs. All channels are factory-calibrated and have a maximum resolution of 16 bits.

All configuration registers and the output values are accessible by a fast and secure point to point connection from the central unit. The on-board controller supports independent

waveform generators, that can be customized by the application developers.

A built-in supervisory mechanism called «Condition Monitoring» can detect, record and report any off-limit usage, such as overvoltage, overcurrent, supply interruption, overtemperature, etc. as well as selected user-defined values.

Configuring the i213 is done easily and time saving by using one of the five methods defined by IEC 61131.

The i213 is designed for harsh rolling stock environments and is fully compliant to EN 50155.

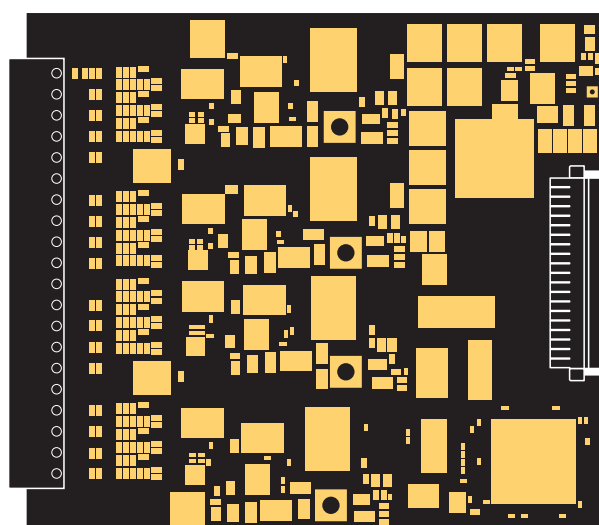
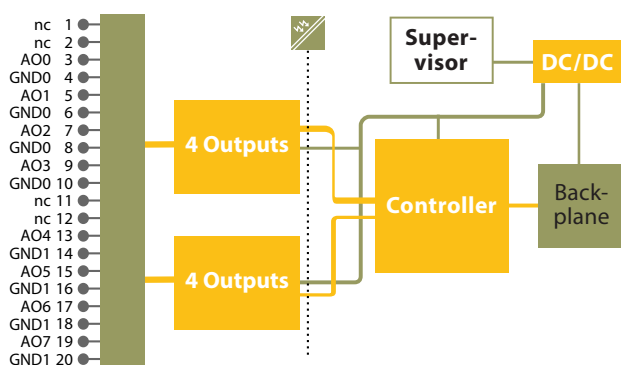
Key Benefits

- User configurable outputs (voltage or current)
- Status LEDs on each channel for easy diagnostics
- Galvanically separated groups for independent outputs
- «Condition Monitoring» sends detailed diagnostics to the central unit
- IEC 61131 for fast and easy programming
- No capacitor replacement necessary
- Complies to EN 50155, EN 50121, IEC 61373

Application Examples

- Driving tachometers
- Driving any kind of analog display
- Generating sinusoidal signals
- Driving sensors with different Voltages
- Controlling any device having an analog control input

i213 Hardware Architecture



Life Cycle Cost

The i213 has been strictly designed without electrolytic capacitors and without any opto-coupler to avoid service cost for changing capacitors and to reduce the life cycle cost of the system. The use of an FPGA with included duagon soft-IP min-

imizes the risk and cost in case of component obsolescence. Any firmware upgrade can be done via the central unit, without the need of removing the device.

Technical Data

Output Channels	– 8 analog output channels divided into two groups – 1000 Vac galvanic isolation between the two groups – Outputs are selectable by application software as – Voltage output: ± 12 Vdc with minimal 1kΩ load resistance – Current output: ± 20 mA with maximum load: 500Ω – Absolute precision: (– 40 °C to + 85 °C) 0.2% of full scale – Output resolution: 16 bits – Max slew rate of each specific channel: 10000 V/s
Configuration	– Programmable by the central unit using IEC 61131 compliant executor and IDE supporting LD, FBD, ST, IL and SFC
Security / Watchdog	– External supervisor for voltage, current and temperature monitoring → «Condition Monitoring» – Communication Link with watchdog supervision
Backplane Communication	– 100 Mbit duagon serial star slave – Secure communication protocol – Configurable status scan cycle time minimum: 1 ms
Diagnostic/ Service	– Configurable diagnostic LEDs for each channel indicating the status of the channel or a user-defined functionality – LEDs for visual status information (power on, running, communicating, failure, I/O, ...) – Diagnostic information readable through the central unit's service interfaces (see i101/i102/i103 for further information about these possibilities)

Power Consumption	– $P_{max} < 6\text{ W}$ – Supports sleep mode – Fully powered via backplane (no external power supply needed) – Interruption Class S2 with ceramic capacitors
Operating Conditions	– Ambient temperature: – 40 to +85 °C (EN 50155, Class TX) – Relative humidity: 75%, max 95% for 30 days per year (conformal coating) EN 60068 – Shock and vibration: According to IEC 61373 category 1, class B – EMI: According to EN 50121 and EN 50155
Dimensions / Weight	– 3TE × 92 × 86 mm, fitting the ionia™ housing – <120 g
Cabling	– Connector type: FMC 1,5/ 20-STF-3,5 or similar – Wire cross section: 0.2mm² to 1.5mm²
Environment	– Fully compliant with RoHS and REACH

d-006957-013238

Order Code and HW Options

Product Type: **i213**

Related Documents

Data Sheet i213  d-006958-nnnnnn.pdf
available at www.duagon.com

**Configure
ionia™ online:**



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